

Research Spike: Dataset Publication Methods for Manuscript Word-Position Data

Track B (Evaluation Methods)

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Objective & Scope

Survey how existing document analysis / computer vision datasets (particularly those involving text localization, layout analysis, or handwritten text) structure their released data and report results, so we can align our own manuscript word-position dataset with community conventions ahead of publication.

Specifically, answering the questions:

- What metrics are used for word/line/region detection tasks (IoU thresholds, mAP, precision/recall/F-measure)?
- Are there metrics specific to historical/manuscript materials (e.g., pixel-level F-measure for binarization, Hausdorff distance for baseline detection)?
- How do datasets define "correct" localization — exact match vs. overlap threshold?
- Do published datasets include a baseline model/script for reproducing scores?
- Is there a standard train/test evaluation protocol competitions use (ICDAR-style) that we should mirror?

Context

We're preparing to publish a dataset centered on word-level bounding box/position annotations on manuscript page images. Before finalizing our own schema, we want a clear picture of the landscape: what formats are standard, what's expected for reproducibility, and what reviewers/users of such datasets tend to expect.

Output

A short comparison document (table + notes) covering ~6-10 datasets, with:

- File/format inventory per dataset
- Evaluation metric(s) used
- Notes on what seemed well-designed vs. what caused friction for users
- A final "recommendations" section proposing what our dataset should include, flagged as
- open questions for the team to decide
- Bibliography of sources provided; citations following Chicago style

Findings:

Methodology

Nine datasets/competitions were selected to span both historical/manuscript document analysis and broader CV/text-recognition challenges.

Historical/manuscript:

- ICDAR cBAD (2017, 2019) – baseline detection
- CLaMM (ICFHR2016, ICDAR2017, ICDAR2019/HDRC-IR) – script and date classification
- IAM Handwriting Database – offline HTR
- DIBCO/H-DIBCO – binarization
- GT4HistOCR – OCR training corpus

Broader CV/text challenges:

- ICDAR Robust Reading Competition (2013, 2015) – scene text
- COCO-Text – scene text in natural images
- PubLayNet – document layout analysis
- FUNSD – form understanding

Dataset & Evaluation Metric Comparison Table Schema

Dataset / Year	Task Type	Primary Metric(s)	Correctness Definition	Included baseline model/script?
ICDAR cBAD, 2017 & 2019	Baseline (text-line) detection in archival documents	Precision, recall, and harmonic F-value, computed line-by-line after greedy matching of predicted to ground-truth baselines (Grüning et al. 2019; Diem et al. 2017, 2019)	No fixed IoU. Each ground-truth baseline gets a tolerance band sized to local x-height/interline spacing; recall is the fraction of that band covered	Yes – official Java evaluator (Transkribus GitHub)
CLaMM, ICFHR2016 / ICDAR2017 / ICDAR2019 (→ HDRCIR)	Script-family and date classification of Latin manuscript pages (2019 adds	Top-1 classification accuracy (2016 Task 1); fuzzy weighted score 0-4 (2016 Task 2); accuracy/confusion matrix (2017	Exact class match for accuracy tasks; partial credit for near-miss classes in the fuzzy task	No public baseline script found – only available via download/registration on clamm.irht.cnrs.fr

	writer/script retrieval)	dating); mAP for retrieval (2019)		
IAM Handwriting Database	Offline handwritten text recognition (HTR)	Character Error Rate (CER) and Word Error Rate (WER), i.e. normalized edit distance (Marti and Bunke 2002)	Not localization-based; scored on transcription accuracy only	No official script form IAM; community defaults to the unofficial “Achen split” (OpenSLR-56)
DIBCO / H-DIBCO, 2009-present	Historical document image binarization	F-measure, pseudo-F-measure (Fps), PSNR, and Distance-Reciprocal Distortion (DRD) (Gatos, Ntirogiannis, and Pratikakis 2009; Ntirogiannis, Gatos, and Pratikakis 2013)	Continuous pixel-level comparison to ground-truth binary image; pseudo-measures weight boundary pixels by stroke width, no single pass/fail cutoff	Yes - DibcoEvaluation tool (GitHub, kzagoris)
GT4HistOCR	Ground-truth training corpus for Fraktur / early modern Latin OCR	None formal; downstream users informally report CER after training a recognizer (Springmann et al. 2018)	N/A - not a competition; no standardized correctness threshold defined	None - is corpus only
ICDAR Robust Reading Competition, 2013 & 2015	Scene text detection, recognition, and end-to-end spotting	2013: DetEval dual-threshold area precision/recall/Hmean (Wolf and Jolion 2006; Karatzas et al. 2013) 2015: single IoU ≥ 0.5 plus edit-distance text match (Karatzas et al. 2015)	2013: matched via $tr = 0.8$ (recall) / $tp = 0.4$ (precision) area thresholds, allowing split/merge matches 2015: strict IoU $\geq 50\%$, Pascal-VOC style	Yes - DetEval toolkit (2013) and RRC portal scripts (2015)
COCO-Text	Scene text detection/recognition in	Precision/Recall/F-score at IoU ≥ 0.5 , explicitly adopting	Single IoU ≥ 0.5 threshold, same as ICDAR2015	Yes - evaluation API (GitHub, andreasveit/coco-t

	MS-COCO natural images	the ICDAR2015 protocol (Veit et al. 2016)		ext)
PubLayNet	Document layout region detection (title, text, list, table, figure)	COCO-style mean Average Precision (mAP), averaged over IoU 0.50-0.95 in steps of 0.05 (Zhong, Tang, and Yepes 2019)	Ten IoU thresholds averaged rather than one cutoff – rewards tighter boxes than a single-threshold protocol	Yes - Faster R-CNN / Mask R-CNN baselines and code (GitHub, ibmaur-nlp)
FUNSD	Word detection, OCR, entity labelling, and entity linking on scanned forms	Word detection: Precision/Recall/F1 at IoU ≥ 0.5 OCR: Levenshtein similarity Entities: F1 Linking: Precision/recall/F1 (Jaume, Ekenel, and Thiran 2019)	Word-level IoU ≥ 0.5 match required before OCR/label/link scoring applies	Yes - faster R-CNN baseline (maskrcnn-benchmark); no dedicated eval API

Metric Deep-Dive Sections

Detection/Localization Metrics

For the non-manuscript datasets, object-detection-style overlap metrics were widely used; however, the specific implementation varies.

ICDAR2013 used the DetEval protocol (Wolf and Jolion 2006), a dual-threshold area-based precision/recall ($tr = 0.8$, $tp = 0.4$) that tolerates one-to-many and many-to-one matches; this is useful for text where split/merge boxes are common.

ICDAR2015 replaced this with a single, stricter IoU ≥ 0.5 threshold “similar to standard practice in object recognition and Pascal VOC” (Karatzas et al. 2015), and COCO-Text explicitly adopted that same simplified protocol (Veit et al. 2016).

PubLayNet goes further, averaging mAP across ten IoU thresholds from 0.50 to 0.95 (Zhong, Tang, and Yepes 2019), which rewards tighter, more precise boxes than a single cutoff would.

FUNSD applies the simplest version, with flat Precision/Recall/F1 at IoU ≥ 0.5 (Jaume, Ekenel, and Thiran 2019).

Manuscript baseline detection breaks from this family entirely: cBAD scores line-level tolerance-band coverage rather than box overlap, because text lines in archival documents are curved, skewed, and variable in height in ways that a rectangular IoU poorly represents (Grüning et al. 2019).

Manuscript/Historical-Specific Metrics

Several manuscript-specific tasks required metrics invented specifically because generic detection metrics didn't fit.

DIBCO/H-DIBCO's pseudo-F-measure weights recall and precision by a distance-to-skeleton / stroke-width function rather than raw pixel overlap, so that a binarization result is credited for following the "spirit" of a handwritten stroke rather than penalized for reasonable edge variation (Ntirogiannis, Gatos, and Pratikakis 2013). Similarly, DRD (Distance-Reciprocal Distortion) similarly weights binarization errors by their visual/perceptual impact rather than treating every mismatched pixel equally. cBAD's tolerance-band matching for baseline detection is the line-detection analogue: it sizes the acceptable margin of error to each line's own x-height and interline spacing instead of one fixed threshold for the whole page (Grüning et al. 2019).

CLaMM sits at the other extreme – because its task is page-level script or date attribution, rather than spatial localization, it dispenses with detection metrics altogether in favor of classification accuracy (with partial credit for near-miss classes in its 2016 fuzzy task) (Cloppet et al. 2016).

Defining “Correct” Localization

Across the surveyed datasets, “correct” ranges from rigid to continuous.

At the rigid end: ICDAR2015, COCO-Text, FUNSD, and PubLayNet all require boxes to clear an IoU threshold (0.5, or 0.5-0.95 averaged) before a detection counts at all – a binary pass/fail.

In the middle: cBAD's tolerance band is still a threshold, but one computed per-line from local geometry rather than fixed globally.

At the continuous end: DIBCO/H-DIBCO's pixel-level scores (PSNR, DRD, pseudo-F) have no single cutoff– every prediction gets a graded score. CLaMM's classification tasks are binary-correct/incorrect at the class level, with the 2016 fuzzy task as a partial exception. This matters for our dataset because manuscript words often have ambiguous or touching boundaries (ligatures, abbreviations, marginal glosses); a rigid single-IoU threshold may unfairly zero out otherwise-reasonable predictions. Follow-on literature critiquing exactly this– TIOU (Liu et al. 2019) and CLEval (Baek et al. 2020) — proposes tightness-aware or character-level alternatives to the single-IoU convention, precisely because split/merge and boundary-tightness errors are punished disproportionately under Pascal-VOC-style matching.

Baseline Models & Reproducibility

Baseline/script availability correlates with how recently and how competition-oriented a dataset is.

Well-supported:

- PubLayNet ships Faster R-CNN and Mask R-CNN baselines with code
- DIBCO/H-DIBCO has a maintained DibcoEvaluation tool
- cBAD has an official Transkribus evaluator
- The ICDAR RRC and COCO-Text both provide portal-hosted or API-based scoring scripts
- FUNSD provides a Faster R-CNN baseline build on maskrcnn-benchmark

Gaps:

- IAM has no official baseline or scoring script from the database maintainers themselves– the community-standard CER/WER protocol and “Aachen split” emerged organically outside the original release (OpenSLR-56). CLaMM’s organizers published competition papers with results tables but no publicly released scoring script was found.
- GT4HistOCR is the starkest gap: it is a ground-truth training corpus, not a benchmark, and defines no evaluation protocol, baseline, or script at all (Springmann et al. 2018).

Standard Evaluation Protocols

- True ICDAR-style protocols– public training data, a held-out or hidden test set, and centralized scoring by organizers– are used by cBAD, DIBCO/H-DIBCO, the ICDAR RRC and CLaMM.
- A looser pattern– a fixed, fully public split with no hidden leaderboard– is used by PubLayNet (train/dev/test partitioned at the journal level to test generalization to unseen templates), FUNSD (fixed 140/50 train/test), and COCO-Text (train/val only at release, no held-out test set).

Given that our dataset will be a one-time publication rather than a recurring competition, the PubLayNet/FUNSD pattern– publicly released, fixed splits, no hidden test set – is the more directly applicable model; the ICDAR style hidden-test/leaderboard pattern is better suited to datasets meant to host repeat competitions.

Analysis Notes

What was Well-Designed

- Public, versioned evaluation code paired with the dataset; PubLayNet’s COCO-API-compatible mAP scoring plus released Faster/Mask R-CNN baselines, DIBCO/H-DIBCO’s maintained DibcoEvaluation tool, and cBAD’s official Transkribus evaluator made scores independently reproducible rather than self-reported, and gave later researchers a fixed target to beat.
- cBAD’s per-line tolerance band, sized to each line’s own x-height and interline spacing, is a well-fitted adaptation: it avoids penalizing correct detections of curved or skewed historical text lines the way a rigid bounding-box IoU would.

- DIBCO's pseudo-F-measure and DRD are similarly well-adapted. They weight errors by perceptual/stroke-level impact instead of treating every mismatched pixel equally, which produces scores that track human judgements of binarization quality more closely than raw pixel accuracy.
- PubLayNet and FUNSD's fixed, fully public train/dev/test splits (no hidden leaderboard) are straightforward for a one-time dataset release to imitate, since they don't require maintaining an ongoing submission/scoring server.

What Caused Friction

- Single, rigid IoU thresholds (ICDAR2015, COCO-Text, FUNSD, PubLayNet's averaged variant) drew direct criticism in follow-in literature. TIOU (Liu et al. 2019) and CLEval (Baek et al. 2020) both argue that a flat IoU cutoff unfairly zeroes out legitimate split/merge detections and ignores how tight an overlap actually is, rather than just whether it clears the bar.
- Datasets without an official released evaluation script left room for inconsistent reporting: IAM never published its own scoring protocol, so the community converged on an unofficial "Aachen split" (OpenSLR-56) that isn't universally used, making cross-paper CER/WER comparisons less reliable. CLaMM's organizers published results tables but no public scoring script was found, so independent reproduction of its accuracy figures is harder to verify.
- GT4HistOCR defines no evaluation protocol, baseline, or correctness threshold at all— it is a training corpus, not a benchmark— which is a reminder that "dataset" and "evaluation-ready benchmark" are not the same deliverable, and we should be explicit about which one we're publishing.
- FUNSD's word-detection F1 (IoU-based) and the entity-label F1 used by later papers built on FUNSD (e.g., LayoutLM) measure different things but share a name, which has caused inconsistent reporting across the literature— a caution for us to name our metrics unambiguously and document exactly what unit (word, line, region) each one stores.

Gaps for Digital-Humanities-Specific Concerns

- None of the surveyed metrics evaluate the accuracy or quality of paleographic/codicological metadata (script type, dating attribution, scribal hand) as a spatially-linked annotation— CLaMM scores script/date classification at the page level only, decoupled from any word- or region-level position data.
- No surveyed dataset defines a metric for damaged, illegible, or marginal content (e.g. marginalia vs. main text, abbreviations, ligatures) — existing bounding-box and pixel metrics are silent on how to score or even flag these cases, which are common in manuscript material and directly relevant to our dataset's DH annotation layer.

Recommendations & Open Questions

Primary localization metric: Adopt word-level Precision/Recall/F1 at $\text{IoU} \leq 0.5$, matching the COCO-Text/FUNSD convention, as our baseline reported metric.

- Open question: Do we additionally report PubLayNet-style mAP averaged over $\text{IoU} 0.50\text{-}0.95$, given manuscript word boxes are typically more regular/consistent than natural scene text and could reasonably support a stricter standard?

Evaluation protocol: Follow the PubLayNet/FUNSD pattern– a fixed, fully public train/val/test split with no hidden leaderboard– since this is a one-time dataset publication rather than a recurring competition

- Open question: Should we still reserve a held-out test subset (not publicly released) to allow future re-benchmarking or a follow-on shared task, ICDAR-style, even without an ongoing competition?

Baseline model/script: Release a public evaluation script (ideally COCO-API-compatible, per PubLayNet) at minimum; a reference baseline detector, if feasible, would match the reproducibility bar set by PubLayNet, DIBCO, cBAD, and FUNSD.

- Open question: Do we have the resources to train and validate a baseline detector before publication, or should v1 ship with the evaluation script only and add a baseline later?

Correctness/tolerance definition: Given manuscript-specific boundary ambiguity (ligatures, abbreviations, marginal glosses, damaged parchment), consider a tolerance-band or relaxed-matching approach in the spirit of cBAD rather than a single rigid IoU cutoff for word boxes

- Open question: Do we adopt a tightness-aware alternative (T_{IoU}- or CLEval-style) outright, or keep plain $\text{IoU} \geq 0.5$ for simplicity and revisit only if reviewers flag it?

DH-specific supplementary metrics: Define at least one metric or reporting convention for the annotation layer beyond bounding-boxes – e.g. inter-annotator agreement or accuracy for script/date attribution where included, and an explicit flag/metric for damaged or ambiguous regions.

- Open question: Is script/date classification an evaluable task in our release (with its own scored metric, CLaMM-style), or purely descriptive metadata attached to each region without a formal correctness measure?

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